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# First-principles study of the effect of the corrosive environmental components on the grain boundary strength in nickel

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The destruction of nickel (Ni)-based alloys occurs mainly along the grain boundaries (GBs) that have been exposed to a corrosive environment. Understanding the mechanism of such exposure at the atomic level is investigated in this study. Based on density functional theory calculations, the GB energy and the GB strength of the  $\Sigma 3(111)$  coherent twin,  $\Sigma 3(1\bar{1}2)$  incoherent twin, and high-angle  $\Sigma 5(0\bar{2}1)$  GBs in Ni are estimated. The change in the GB strength in Ni is considered upon the change in the concentration of corrosive elements, such as O, Cl, and S. The highest GB strength is found for the GB in the O-rich environment, while the lowest GB strength is found for the GB in the Cl-rich environment. In the case of interstitial O and Cl impurities, the energy that affects the GB's strength or brittleness comes both from the mechanical factor, which is related to the atomic displacements around the impurity, and from the chemical factor, which is related to the bond formation/breaking around the impurity. The results of this study not only deepen the fundamental understanding of the impurity effect on the GB strength in Ni and Ni-based alloys, but also provide valuable information for production and technological application of these alloys.

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## Introduction

High corrosion-resistant superalloys, such as Ni-based alloys, have become primary candidate materials for structural components under extreme operating conditions. For example, Ni-based alloys are widely used as structural and welding materials in nuclear reactors due to their excellent mechanical properties and corrosion resistance at high temperatures. Alloys, such as Inconel 718, are used for high-temperature engine components, such as turbine blades, due to their excellent resistance to both heat and corrosive gases.<sup>1</sup> Alloys, such as Hastelloy, are utilized in the manufacturing and installation of marine industrial equipment in marine environments due to their strong salt corrosion resistance.<sup>2</sup> On the other hand, the corrosion resistance of alloys in an aggressive environment, including Ni-based alloys, has a decisive influence on the service life, safety, and reliability of these alloys.

The corrosion environment for Ni-based alloys can be classified based on the type of corrosion occurring, such as

uniform, localized, or stress corrosion cracking and the temperature conditions involved, such as aqueous, high-temperature gas, or molten salts.<sup>3</sup> Certain types of corrosion are associated with the presence of certain chemical species, including oxidizing agents, such as chlorine gas, aggressive acids, or conditions that are reducing in nature. Specifically, salt-vapor environments are often harmful to Ni-based alloys and the intensity of their corrosion increases when salt vapor is mixed with other salts.<sup>4</sup>

It has been previously shown that the corrosion rate of Ni-based alloys increases when they are simultaneously exposed to sulfur- and chlorine-rich environments.<sup>5</sup> Experimental studies have shown that in environments containing sulfide gases, sulfur compounds can damage the oxide layer on the surface of NiCrMoAl, causing significant corrosion<sup>6</sup>. In sulfide salt solutions, the corrosion of the Incoloy 825 alloy becomes more severe as the sulfur content increases.<sup>7</sup> Based on electrochemical tests and surface analysis, it is revealed that beneath the deposits of elemental sulfur, a strong acidic environment is formed due to the hydrolysis of elemental sulfur. This acidic condition greatly degenerates the passive film, resulting in serious corrosion. In a molten salt environment, a higher concentration of chloride ions at the alloy-salt interface greatly accelerates the corrosion of the alloy.<sup>8,9</sup> The presence of chloride and sulphate eutectic salts in such environments further increases the corrosion rate.<sup>10</sup> In supercritical water environments, certain supercritical water

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containing phosphate, sulfate, chloride and oxygen,<sup>11</sup> certain atmospheric gases and noble gases,<sup>12</sup> the corrosion of Ni-based alloys is exacerbated.

Another critical issue related to Ni-based alloys is stress corrosion cracking, which involves cracks growing along GBs due to a combination of tensile stress and a corrosive environment. In this sense, GBs represent an important class of two-dimensional extended defects, and the macroscopic strength of polycrystalline materials depends significantly on the GB cohesion. Nickel-based alloy processing aims to manage GB formation through GB engineering. This method improves corrosion resistance by increasing the amount of beneficial, low-energy coincidence site lattice boundaries and decreasing the overall connectivity of the GB network. This promotes the development of specific coincidence site lattice boundaries such as  $\Sigma 3$  GBs, which exhibit weaker resistance to dislocation slip and can improve ductility, and result in smaller grain sizes.<sup>13</sup> This strategy enhances resistance against different types of corrosion, including intergranular corrosion, by forming a more stable protective layer and preventing cracks from spreading along the GBs.<sup>14,15</sup>

It has long been recognized that intergranular embrittlement, usually associated with segregation of impurities on the GB, can lead to a dramatic reduction of the ductility and strength of alloys.<sup>16</sup> In alumina, it has been demonstrated that different impurities have distinct effects on the GB structures, primarily determined by the ionic size mismatch between the impurities and host cations.<sup>17</sup> Elements, such as B, C, N, O, P, and S, can enhance the stabilization of the  $\Sigma 5$  (310) GB in iron, while H and Si impurities lead to the opposite effect.<sup>18</sup> The investigation of sulfur induced embrittlement of the  $\Sigma 5(012)$  GB in Ni has revealed the extension of the Ni GB as a result of the repulsion of the segregated and neighboring sulfur atoms.<sup>19</sup> It has been shown that the abnormal grain growth may be a result of the highly anisotropic nature of complex formation in S-doped Ni.<sup>20</sup> The oxidation of the  $\Sigma 3(111)$  coherent twin,  $\Sigma 3(112)$  incoherent twin, and  $\Sigma 5(210)$  high-angle GB in Ni is also deeply understood.<sup>21</sup> Particularly, it has been shown that the Ni/NiO metal-oxide interface can be very weak and serve as a favorable site for crack initiation.

On the computational side, the most accepted and accurate approach to study the segregation and GBs in metals and alloys at the atomistic level is density functional theory (DFT).<sup>22</sup> First-principles calculations have been used to develop a method for quick estimation of the GB energy at 0 K for seven bcc metals, namely V, Cr, Fe, Nb, Mo, Ta, and W.<sup>23</sup> Based on the results of the DFT simulations, one of the largest databases of GB properties has been created, containing 327 GBs of 58 elemental metals.<sup>24</sup> To define the mechanism of the fracture of 69 binary Al alloys due to GB segregation, multiple fracture paths have been evaluated utilizing all-electron first-principles calculations.<sup>25</sup> High-throughput first-principles calculations have been implemented to calculate GB segregation energies and GB-strengthening energies of most solutes in many technologically relevant metals including Al, Cu, Fe, Mg, Mo, Nb, Ni, Ta, Ti, and W.<sup>26</sup> Self-diffusion in the symmetric tilt GBs (112) and (210) in bcc iron taking into account transfer to the ferromagnetic state has been

investigated at the atomic scale using a DFT approach.<sup>27</sup> First-principles calculations have also been effectively used in studying the adsorption and segregation of the corrosion adsorbate in the GB of various metals.<sup>28</sup>

Despite many above-mentioned important individual studies on the GB strength change due to the impurity segregation, there is still a need for a thorough understanding at the atomic level and a comprehensive comparison of how corrosive impurities, such as Cl, O and S affect the strength of various GBs in Ni and Ni-based alloys. This understanding is essential for developing and improving high-strength Ni-based alloys and other materials.

In this study, the extent and mechanisms of GB strength reduction in Ni due to the presence of various corrosive elements, such as Cl, O, and S, are systematically investigated based on first-principles simulations. The concentration of the Cl, O, and S ions varies to consider stages from partially to fully occupied GBs by these impurities. In Ni, GBs are classified based on their structure and misorientation angle. Low-angle GBs are typically more stable and have lower energy than high-angle GBs. Special GBs with low  $\Sigma$  values, such as  $\Sigma 3$ ,  $\Sigma 5$ , and  $\Sigma 11$ , have a more ordered structure and can exhibit different properties compared to general high-angle GBs.<sup>29</sup> Although experimental works specify the  $\Sigma 3(111)$  coherent twin as being dominant in Ni, the  $\Sigma 3(1\bar{1}\bar{2})$  incoherent twin boundary and  $\Sigma 5(0\bar{2}1)$  are also considered in this study to represent the non-ideal twin boundaries and high-angle GBs, respectively.<sup>30,31</sup>

## Methods

### Calculation details

In this work, DFT-based calculations were performed using the Vienna *Ab initio* Simulation Package.<sup>32</sup> The exchange–correlation energies of electrons were described using the generalized gradient approximation in the Perdew–Burke–Ernzerhof functional.<sup>33</sup> The interaction between ionic cores and valence electrons was accounted for using the Projector-Augmented-Wave method.<sup>34</sup> In all calculations, system optimization was performed using the conjugate gradient algorithm. Spin polarization was included to account for magnetic effects. Due to the ferromagnetic nature of Ni, to ensure the correct orientation of spins in the calculation, each Ni atom was assigned an initial magnetic moment of  $5\mu_B$  (a starting MAGMOM value equal to 5.0), while magnetic moments of other atoms were set to  $0.6\mu_B$  (a starting MAGMOM value equal to 0.6). During the optimization process, the convergence criteria for residual force and energy were set to  $0.05 \text{ eV } \text{\AA}^{-1}$  and  $10^{-5} \text{ eV}$ , respectively. For electronic self-consistent simulations, the total energy convergence accuracy criterion was set to  $10^{-8} \text{ eV}$ . The cutoff energy was set to 520 eV. The Brillouin zone was sampled with a  $1 \times 4 \times 7$ ,  $2 \times 3 \times 7$ , and  $5 \times 4 \times 2$   $k$ -mesh grid using the Monkhorst–Pack scheme for the cases when the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}\bar{2})$ , and  $\Sigma 5(0\bar{2}1)$  GBs were simulated, respectively. The simulation parameters were chosen to maintain a balance between accuracy and computational cost of the simulations

and based on similar previous studies.<sup>21</sup> The Grimme's DFT-D3 method was used to accurately describe long-range van der Waals interactions between corrosive impurities and Ni atoms.<sup>35</sup> The charge differential difference (CDD) analysis was utilized to quantify and visualize the charge transferred between corrosive impurities and Ni atoms. The amount of the charge transferred between the impurities and the GB atoms was estimated using the Bader charge analysis.<sup>36,37</sup> The atomic structures were visualized using the VESTA 3 program.<sup>38</sup>

### GB model

There are different approaches to calculate the GB energy: the fixed grain distance setup, the cell relaxation setup and the vacuum on top setup.<sup>39</sup> The fixed grain distance setup was used in this work. Models for the considered GBs were created based on those available at Crystalum.<sup>24</sup> The total numbers of atoms in the  $\Sigma 3(111)$  coherent twin,  $\Sigma 3(1\bar{1}\bar{2})$  incoherent twin, and high-angle  $\Sigma 5(0\bar{2}1)$  GB models were 96, 88, and 80, respectively. Their dimensions after structure optimization were:  $8.51 \text{ \AA} \times 4.92 \text{ \AA} \times 24.08 \text{ \AA}$ ,  $4.92 \text{ \AA} \times 12.04 \text{ \AA} \times 17.03 \text{ \AA}$ , and  $6.95 \text{ \AA} \times 7.77 \text{ \AA} \times 15.54 \text{ \AA}$ , respectively.

The free surface energy  $\gamma_{\text{FS}}$  which is generally defined as the work required to build a unit of area of a given surface was calculated as follows:

$$\gamma_{\text{FS}} = \frac{E_{\text{FS}} - \frac{N_{\text{FS}}}{N_{\text{bulk}}} E_{\text{bulk}}}{2A_{\text{FS}}}, \quad (1)$$

where  $N_{\text{FS}}$  and  $N_{\text{bulk}}$  are the number of atoms in the corresponding free surface structure and the bulk structure, respectively.  $E_{\text{bulk}}$  is the total energy of the supercell of the bulk (Fig. 1a, e and h),  $E_{\text{FS}}$  is the total energy of the supercell of the surface (Fig. 1b, f and i), and  $A_{\text{FS}}$  is the area of the surface.

The GB energy  $\gamma_{\text{GB}}$  was calculated as follows:

$$\gamma_{\text{GB}} = \frac{E_{\text{GB}} - \frac{N_{\text{GB}}}{N_{\text{bulk}}} E_{\text{bulk}}}{2A_{\text{GB}}}, \quad (2)$$

where  $A_{\text{GB}}$  is the area of the GB and  $E_{\text{GB}}$  is the total energy of the supercell with the GB (Fig. 1c, g and j).

The work of separation  $W_{\text{sep}}$  which is the energy needed to separate two grains was calculated as follows:<sup>40</sup>

$$W_{\text{sep}} = 2\gamma_{\text{FS}} - \gamma_{\text{GB}}, \quad (3)$$

and the GB strength was calculated as follows:

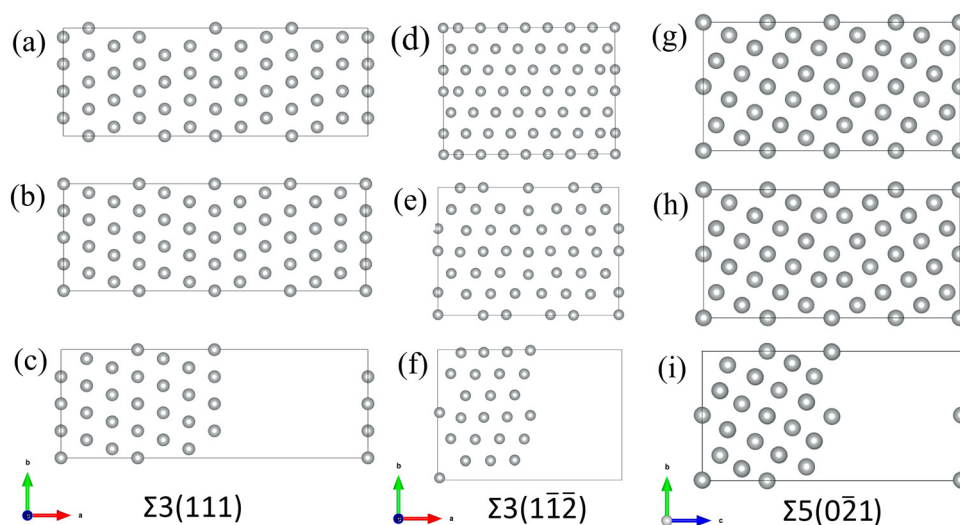
$$\begin{aligned} \gamma^{\text{imp}} &= W_{\text{sep}} + \frac{[(E_{\text{FS}}^{\text{oxygen}} - E_{\text{FS}}^{\text{pure}}) - (E_{\text{GB}}^{\text{oxygen}} - E_{\text{GB}}^{\text{pure}})]}{A_{\text{GB}}} \\ &= W_{\text{sep}} + \Delta\gamma^{\text{imp}}, \end{aligned} \quad (4)$$

where  $E_{\text{FS}}^{\text{oxygen}}$  is the total energy of a free surface system containing surface oxygen (sulfur or chlorine) atoms, and  $E_{\text{FS}}^{\text{pure}}$  is the total energy of the same system without oxygen (sulfur or chlorine) atoms. Similar meanings can be applied to  $E_{\text{GB}}^{\text{oxygen}}$  and  $E_{\text{GB}}^{\text{pure}}$ .

## Results and discussion

To create surface and GB models of Ni, we first created several bulk structures of different sizes and optimized them (Fig. 1a, d and g). The size of these bulk structures is chosen so that the subsequently created models of the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}\bar{2})$ , and  $\Sigma 5(0\bar{2}1)$  GBs are of appropriate size. The calculated lattice constant of the face-centered cubic (FCC) Ni structure was  $3.48 \text{ \AA}$ , which is close to the experimental and theoretical values ( $3.48\text{--}3.52 \text{ \AA}$ ) reported in the literature.<sup>41</sup>

$\gamma_{\text{FS}}$  which characterizes the excess energy on the surface of a material compared to its volume is evaluated. The main point is



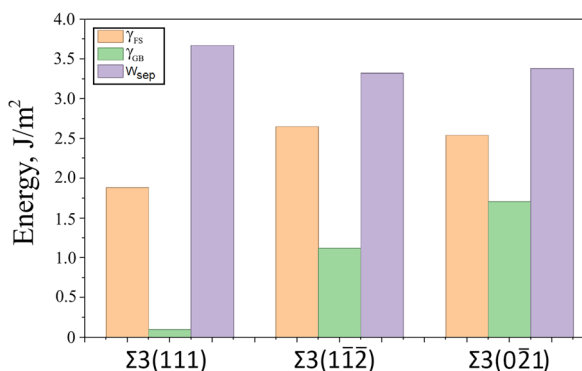
**Fig. 1** Models showing the optimized bulk structures of Ni (a), (d), and (g). Models showing the optimized structures of the  $\Sigma 3(111)$  coherent twin GB (b), the  $\Sigma 3(1\bar{1}\bar{2})$  incoherent twin GB (e), and  $\Sigma 5(0\bar{2}1)$  GB (h). Models showing the optimized surface structures, where (c) corresponds to (b), (f) corresponds to (e), and (i) corresponds to (h).

**Table 1**  $\gamma_{\text{FS}}$  ( $\text{J m}^{-2}$ ),  $\gamma_{\text{GB}}$  ( $\text{J m}^{-2}$ ) and  $W_{\text{sep}}$  ( $\text{J m}^{-2}$ ) of Ni GBs

| GB type              | $\Sigma 3(111)$ | $\Sigma 3(1\bar{1}\bar{2})$ | $\Sigma 5(0\bar{2}1)$ |
|----------------------|-----------------|-----------------------------|-----------------------|
| $\gamma_{\text{FS}}$ | 1.88            | 2.65                        | 2.54                  |
| $\gamma_{\text{GB}}$ | 0.10            | 1.12                        | 1.71                  |
| $W_{\text{sep}}$     | 3.67            | 3.32                        | 3.38                  |

that the higher the surface energy, the more actively the surface interacts with other substances, for example, with liquids (wettability) or with other solids (adhesion). GBs can influence atomic arrangement at the surface, affecting the  $\gamma_{\text{FS}}$ . For nanocrystalline materials such as Ni, the  $\gamma_{\text{FS}}$  is inversely proportional to the GB density, and as the density of GBs increases, the  $\gamma_{\text{FS}}$  decreases.<sup>42</sup> Thus, by calculating the  $\gamma_{\text{FS}}$  of Ni surfaces containing various GBs, it is possible to estimate their activity towards corrosive environmental components. Table 1 and Fig. 2 show  $\gamma_{\text{FS}}$  for the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}\bar{2})$ , and  $\Sigma 5(0\bar{2}1)$  GBs in Ni calculated following eqn (1). The surface containing the  $\Sigma 3(1\bar{1}\bar{2})$  GB has the highest  $\gamma_{\text{FS}}$  value of  $2.65 \text{ J m}^{-2}$ , while the surface containing the  $\Sigma 3(111)$  GB has the lowest  $\gamma_{\text{FS}}$  value of  $1.88 \text{ J m}^{-2}$ . Therefore, the presence of the  $\Sigma 3(1\bar{1}\bar{2})$  GB in Ni significantly enhances its reactivity towards corrosive environmental components. Notably, the calculated  $\gamma_{\text{FS}}$  values are in good agreement with those reported in ref. 43, confirming the reliability of our simulation models.

The corrosion processes preferentially occur in the GBs due to their high energy compared to the bulk. This is because the existence of the GBs in metals, specifically in Ni, causes a deviation from the ideal lattice structure due to the misfit between atoms across the GB.<sup>44,45</sup> Such local distortions lead to an increase of the entropy and therefore the free energy of that region. The higher the misorientation of the GB, the higher the energy increase. In turn, the higher the  $\gamma_{\text{GB}}$ , the less stable they are and they tend to transform to lower energy configurations.<sup>46</sup> The  $\gamma_{\text{GB}}$  values for the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}\bar{2})$ , and  $\Sigma 5(0\bar{2}1)$  GBs in Ni, calculated using eqn (2), are collected in Table 1 and shown in Fig. 2. In accordance with experimental observations,<sup>31</sup> the  $\Sigma 3(111)$  GB has the lowest  $\gamma_{\text{GB}}$  value of  $\sim 0.10 \text{ J m}^{-2}$ , being the most favorable GB configuration in Ni. The  $\Sigma 3(1\bar{1}\bar{2})$  exhibits a  $\gamma_{\text{GB}}$  value of  $1.12 \text{ J m}^{-2}$ , while the  $\Sigma 5(0\bar{2}1)$  GB exhibits a  $\gamma_{\text{GB}}$  value of  $1.71 \text{ J m}^{-2}$ . This result is consistent with the common notion

**Fig. 2**  $\gamma_{\text{FS}}$  (orange),  $\gamma_{\text{GB}}$  (green), and  $W_{\text{sep}}$  (violet) for the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}\bar{2})$ , and  $\Sigma 5(0\bar{2}1)$  GBs.

that high angle GBs in Ni, such as the  $\Sigma 5(0\bar{2}1)$  GB, have the highest  $\gamma_{\text{GB}}$ . Notably, the  $\gamma_{\text{FS}}$  is higher than  $\gamma_{\text{GB}}$  since the surface atoms have fewer neighboring atoms to bond to compared to atoms at GBs thus the surface atoms have more unpassivated bonds as a result of local lattice distortions.

The  $W_{\text{sep}}$  is a crucial factor in determining a material's mechanical properties, such as its fracture toughness and resistance to intergranular failure.<sup>47</sup> One of the key factors relative to the  $W_{\text{sep}}$  is the atomic arrangement and misorientation at the GB, which affect the strength of the GB. The changes in the  $W_{\text{sep}}$  can be immediately utilized as indicators of the GB cohesive strength: the lower the misorientation at the GB, the lower the GB cohesion. The  $W_{\text{sep}}$  values for the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}\bar{2})$ , and  $\Sigma 5(0\bar{2}1)$  GBs in Ni, calculated using eqn (3), are collected in Table 1 and shown in Fig. 2. The  $\Sigma 3(111)$  GB with the highest misorientation exhibits the highest  $W_{\text{sep}} = 3.67 \text{ J m}^{-2}$ , while the  $\Sigma 3(1\bar{1}\bar{2})$  and  $\Sigma 5(0\bar{2}1)$  GBs have lower  $W_{\text{sep}}$  values of  $3.32 \text{ J m}^{-2}$  and  $3.38 \text{ J m}^{-2}$ , respectively. Overall, all considered GBs demonstrate relatively high strength. However, there are other factors that affect the  $W_{\text{sep}}$  and GB strength. In particular, solutes or impurities, which segregate at the GBs, alter their composition and potentially weaken or strengthen them.

The corrosion often initiates from the GB-intersections, where the higher binding energy can help the absorption of corrosive elements and thereby promote the degradation.<sup>48</sup> The binding energy  $E_{\text{b}}$  at the GB-surface intersection for the O, S and Cl ions is calculated as follows:

$$E_{\text{b}} = E_{\text{bulk}} + E_{\text{impurity}} - E_{\text{bulk+impurity}} \quad (5)$$

where  $E_{\text{bulk}}$ ,  $E_{\text{impurity}}$ , and  $E_{\text{bulk+impurity}}$  are the total energies of the GB configuration in Ni without the impurity atom, the total energy of an isolated impurity atom, and the total energy of the same GB configuration in Ni containing an impurity atom, respectively.

For the  $\Sigma 3(111)$  GB in Ni, the calculations reveal the highest  $E_{\text{b}} = 2.55 \text{ eV}$  for the O impurity atom, compared to  $E_{\text{b}} = 0.57 \text{ eV}$  for the S impurity atom, while  $E_{\text{b}} = -3.79 \text{ eV}$  is found for the Cl impurity atom. For the  $\Sigma 3(1\bar{1}\bar{2})$  GB in Ni, the highest  $E_{\text{b}} = 5.12 \text{ eV}$  and  $E_{\text{b}} = 4.89 \text{ eV}$  were obtained for the S and O impurity atoms, respectively, while  $E_{\text{b}} = 1.05 \text{ eV}$  is found for the Cl impurity atom. Similarly, for the  $\Sigma 5(0\bar{2}1)$  GB in Ni, the highest  $E_{\text{b}} = 2.95 \text{ eV}$  and  $E_{\text{b}} = 2.80 \text{ eV}$  were obtained for the O and S impurity atoms, respectively, while  $E_{\text{b}} = -1.83 \text{ eV}$  is found for the Cl impurity atom. Therefore, the calculations show that Cl forms a stronger bond with Ni in the GB-surface intersection than O or S.

Because solutes and impurities that prefer segregation to the GB over the free surface improve GB cohesion upon segregation, whereas solutes that decrease GB cohesion prefer segregation to the free surface, the  $W_{\text{sep}}$  can be used as an indicator of a potential effect of solutes and impurities on the GB cohesive strength. The corrosion of metals and alloys occurs in aggressive environments such as oxygen-enriched atmospheres and various solutions rich in sulfur and chlorine.<sup>49–51</sup> The concentration of ions of O, S and Cl may increase in the GB under various conditions accelerating the corrosion process.



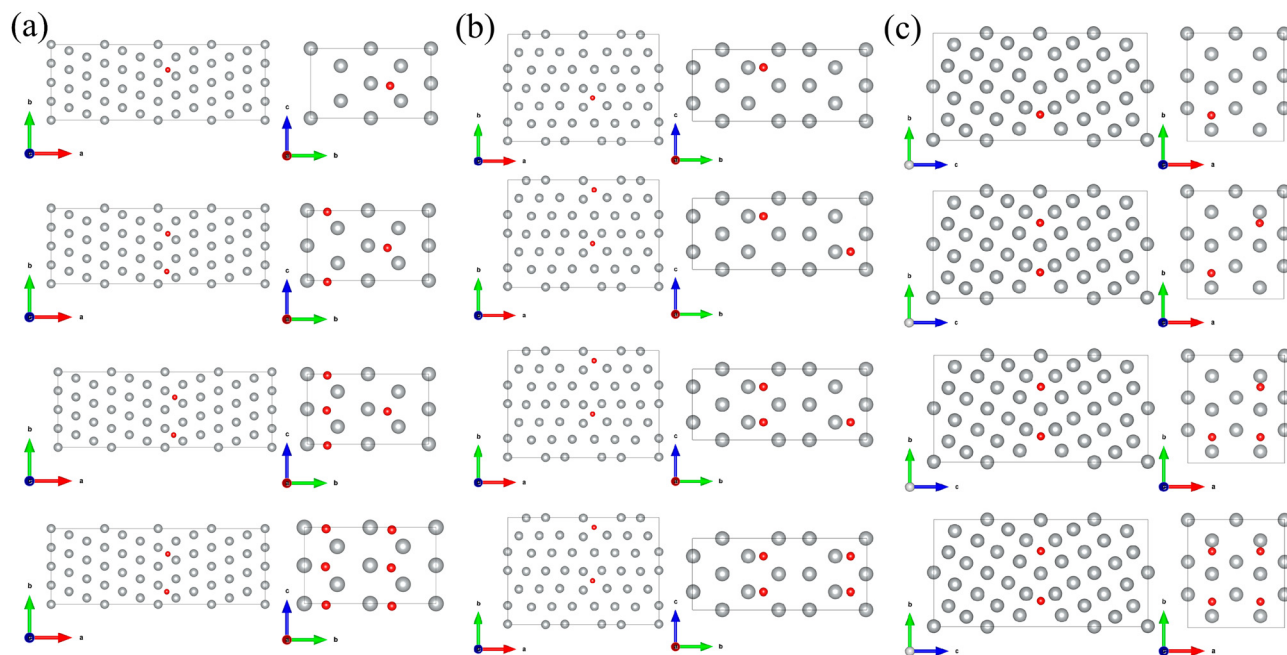


Fig. 3 The O/S/Cl impurity sites (red spheres) in the  $\Sigma 3(111)$  (a),  $\Sigma 3(1\bar{1}2)$  (b), and  $\Sigma 5(0\bar{2}1)$  (c) GBs in Ni (gray spheres).

The effect of the O, S and Cl ions and their concentration on the GB strength in Ni is further discussed. Fig. 3 shows the atomic structures of the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}2)$ , and  $\Sigma 5(0\bar{2}1)$  GBs in Ni with impurities (red spheres) segregated in the interstitial sites at the interface of two grains. Regular lattice sites were not considered as they were found less energetically favorable.<sup>21</sup> Ion concentration is controlled by the number of impurity atoms, ranging from 1 to 4, at the interface of two grains.

Considering all possible impurity sites and ion concentrations as shown in Fig. 3,  $\gamma^{\text{imp}}$  is calculated following eqn (4). The data is collected in Table 2 and depicted in Fig. 4. According to Fig. 4 in all considered cases  $\gamma^{\text{imp}}$  decreases with an increase of the ion concentration. In the case of the  $\Sigma 3(111)$  GB, the highest  $\gamma^{\text{imp}}$ , ranging from  $0.28 \text{ J m}^{-2}$  to  $2.50 \text{ J m}^{-2}$ , was attributed to the O segregation. While the lowest  $\gamma^{\text{imp}}$ , ranging from  $-4.76 \text{ J m}^{-2}$  to  $0.95 \text{ J m}^{-2}$ , was attributed to the Cl segregation. This suggests that even a low concentration of Cl atoms in the  $\Sigma 3(111)$  GB of Ni can significantly reduce its strength, leading to material degradation.

Similarly, a relatively low concentration of the S atoms reduces the strength of the  $\Sigma 3(111)$  GB of Ni. In the case of the  $\Sigma 3(1\bar{1}2)$  GB, the highest  $\gamma^{\text{imp}}$  is attributed to the S atoms segregation ( $\gamma^{\text{imp}}$  ranging from  $2.37 \text{ J m}^{-2}$  to  $3.23 \text{ J m}^{-2}$ ), however, the  $\gamma^{\text{imp}}$  attributed to the O atoms segregation is almost at the same level ( $\gamma^{\text{imp}}$  ranging from  $2.24 \text{ J m}^{-2}$  to  $3.08 \text{ J m}^{-2}$ ). Significantly lower  $\gamma^{\text{imp}}$  is found in the case of Cl atom segregation ( $\gamma^{\text{imp}}$  ranging from  $1.25 \text{ J m}^{-2}$  to  $2.51 \text{ J m}^{-2}$ ). On the other hand, the  $\gamma^{\text{imp}}$  is positive for all impurities, thus, although the  $\Sigma 3(1\bar{1}2)$  GB strength in Ni decreases with increasing impurity concentration at the GB interface, the degradation material is mitigated. In the case of the  $\Sigma 5(0\bar{2}1)$  GB in Ni, the largest  $\gamma^{\text{imp}}$  is also found when O and S atoms are segregated at the GB interface. With increasing the O and S concentration the

$\gamma^{\text{imp}}$  decreases, indicating a mitigation of the degradation of the material. Notably, the lowest  $\gamma^{\text{imp}}$  is observed when Cl atoms are segregated at the GB interface. An increase in the Cl concentration leads to a remarkable decrease in the  $\gamma^{\text{imp}}$  to  $-3.30 \text{ J m}^{-1}$ . This suggests that even a low concentration of Cl atoms in the  $\Sigma 5(0\bar{2}1)$  GB of Ni can significantly reduce its strength, leading to material degradation.

It can be concluded that the  $\Sigma 3(1\bar{1}2)$  GB strength in Ni remains high in the presence of the O, S, and Cl impurities, whereas the strength of the  $\Sigma 3(111)$  and  $\Sigma 5(0\bar{2}1)$  GBs in Ni significantly decreases in the presence of the Cl impurities. Similar observations have been recently reported in experimental studies,<sup>52,53</sup>

Table 2  $\gamma^{\text{imp}}$ ,  $\text{J m}^{-2}$  of the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}2)$ ,  $\Sigma 5(210)$  GBs in Ni depending on the number of interstitial atoms in Ni GBs

| Type of GB                  | $\Sigma 3(111)$ | $\Sigma 3(1\bar{1}2)$ | $\Sigma 5(0\bar{2}1)$ |
|-----------------------------|-----------------|-----------------------|-----------------------|
| Number of impurity O atoms  |                 |                       |                       |
| 1                           | 2.50            | 3.08                  | 2.78                  |
| 2                           | 1.33            | 2.59                  | 2.28                  |
| 3                           | 0.78            | 2.58                  | 1.20                  |
| 4                           | 0.28            | 2.24                  | 0.75                  |
| Number of impurity S atoms  |                 |                       |                       |
| 1                           | 1.96            | 3.23                  | 2.61                  |
| 2                           | 0.34            | 3.00                  | 1.88                  |
| 3                           | -0.32           | 2.89                  | 0.47                  |
| 4                           | -1.20           | 2.37                  | -0.05                 |
| Number of impurity Cl atoms |                 |                       |                       |
| 1                           | 0.95            | 2.51                  | 1.95                  |
| 2                           | -1.67           | 1.57                  | 0.22                  |
| 3                           | -3.17           | 1.16                  | -0.98                 |
| 4                           | -4.76           | 1.25                  | -3.30                 |

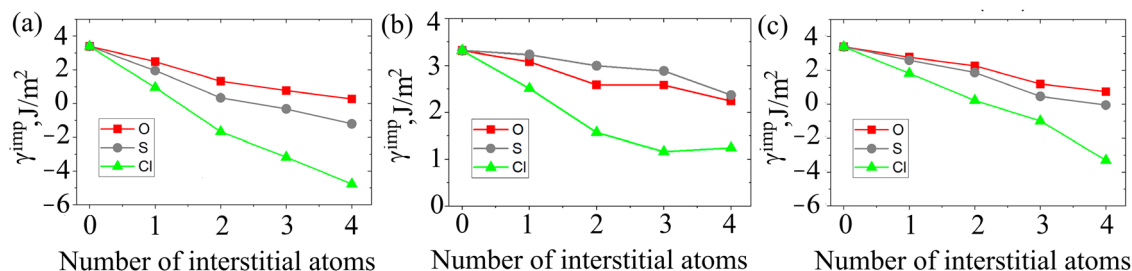


Fig. 4  $\gamma_{\text{imp}}$  of the  $\Sigma 3(111)$  (a),  $\Sigma 3(11\bar{2})$  (b), and  $\Sigma 5(0\bar{2}1)$  (c) GBs in Ni as a function of the number of impurity atoms.

where Cl ions form stable compounds with metal ions promoting localized degradation of the alloy.

To more deeply understand the nature of the  $\Sigma 3(11\bar{2})$  GB strength and a decrease in the strength of the  $\Sigma 3(111)$  and  $\Sigma 5(0\bar{2}1)$  GBs, charge transfer and atomic bonds between impurity atoms and Ni atoms at the GB interface are analyzed. Two most representative cases are considered, when 1 and 4 impurity atoms are segregated in the GB interface. According to Table S1, the average distance between Ni atoms at the GB interface increases with an increase in the O concentration, from 4.24 Å to 4.49 Å in the case of the  $\Sigma 3(111)$  GB, from 4.07 Å to 4.15 Å in the case of the  $\Sigma 3(11\bar{2})$  GB, and from 3.36 Å to 3.59 Å in the case of the  $\Sigma 5(0\bar{2}1)$  GB. An increase of the S concentration at the GB interface also leads to a moderate increase of the average distance between Ni atoms at the interface from 4.22 Å to 4.95 Å in the case of the  $\Sigma 3(111)$  GB, from 4.12 Å to 4.35 Å in the case of the  $\Sigma 3(11\bar{2})$  GB, and from 3.41 Å to 3.69 Å in the case of the  $\Sigma 5(0\bar{2}1)$  GB. Noticeable changes in the average distance between Ni atoms at the GB interface are observed in the case of Cl segregation there. It significantly increases from 4.24 Å to 5.14 Å in the case of the  $\Sigma 3(111)$  GB, from 4.16 Å to 4.73 Å in the case of the  $\Sigma 3(11\bar{2})$  GB, and from 3.43 Å to 3.87 Å in the case of the  $\Sigma 5(0\bar{2}1)$  GB. Chlorine introduces significant changes in the Ni GBs due to the increase in the bond lengths between Ni atoms of neighboring grains. This leads to weakening of the Ni bonds in the GBs and a local decrease in strength at this region of the GB. This is especially noticeable in the example of the  $\Sigma 3(111)$  GB.

Fig. 5a shows that the presence of one impurity O atom at the  $\Sigma 3(111)$  GB leads to a charge redistribution between the Ni and O atoms. According to Bader analysis, O is a strong acceptor in this case, receiving about 1.22  $e$  per atom. Electron localization is observed on the newly formed Ni–O bonds, which indicates partial passivation of Ni bonds in the GB and the formation of strong Ni–O bonds. Fig. 5b shows the case when four O atoms are present in the  $\Sigma 3(111)$  GB. In this scenario, O is also a strong acceptor, but the number of electrons transferred by neighboring Ni atoms to O atoms is slightly lower, 1.16  $e$  per atom. In addition, for each of the O atoms, electron localization is observed on the Ni–O bonds with neighboring Ni1, Ni2, Ni3, and Ni4 atoms, suggesting that the O concentration may be excessive.

Fig. 5c and d presents CDD plots for the  $\Sigma 3(111)$  GB in Ni with 1 and 4 impurity Cl atoms. Bader analysis shows that one

impurity Cl atom accepts up to 0.94  $e$  per atom (Fig. 5c), while in the case of four impurity Cl atoms each atom accepts up to 0.70  $e$  per atom. This is due to a charge redistribution between Cl atoms through neighbouring Ni atoms as shown in Fig. 5d. In the case of one S impurity atom at the  $\Sigma 3(111)$  GB, accumulation of 0.94  $e$  per S atom is found (Fig. 5e).

Electron localization is also observed on the Ni–S bonds with neighboring Ni atoms. The presence of four impurity S atoms in the  $\Sigma 3(111)$  GB induces a charge redistribution between the Ni and S atoms (Fig. 5f). Thus, as in the case of O, the charge accumulation on the S atoms slightly decreases to 0.70  $e$  per atom.

In addition, electron localization is observed on the Ni–Cl bonds confirming formation of these newly formed bonds in the GB. Notably, the highest charge transfer from Ni atoms is found to O atoms, and the lowest to the S atoms, which is reasonable, since the oxygen (3.44) has the highest electronegativity difference with nickel (1.91) and sulfur (2.58) has the lowest electronegativity with nickel among considered impurity elements. And the higher the difference in electronegativity between two atoms the higher the charge transfer between them.

Table 3 collects the data on the average bond lengths and the average charge transfer values between impurity O, S, and

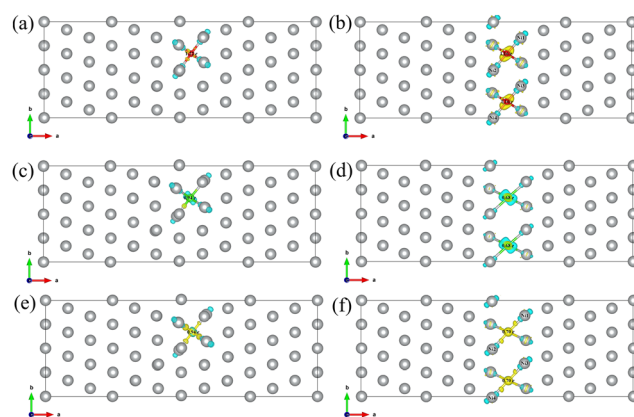


Fig. 5 CDD plots for the  $\Sigma 3(111)$  GB in Ni (grey spheres) with 1 (a) and 4 (b) impurity O (red spheres) atoms, with 1 (c) and 4 (d) impurity Cl (green spheres) atoms, and with 1 (e) and 4 (f) impurity S (yellow spheres) atoms. The yellow/blue color represents an accumulation/depletion of electrons. The isosurface value of 0.015 Å<sup>−3</sup> is adopted.

**Table 3** Average bond lengths and average values of the charge lost by Ni atoms in the GBs

| GB type                        | Average/shortest Ni–O bond length, Å |           | Average charge loss of Ni atom, $e$ per atom |         |
|--------------------------------|--------------------------------------|-----------|----------------------------------------------|---------|
|                                | 1 atom                               | 4 atoms   | 1 atom                                       | 4 atoms |
| $\Sigma 3(111)$ GB             |                                      |           |                                              |         |
| O                              | 1.88/1.87                            | 1.94/1.88 | 0.19                                         | 0.3     |
| S                              | 2.00/1.99                            | 2.10/2.07 | 0.12                                         | 0.18    |
| Cl                             | 2.02/2.01                            | 2.18/2.14 | 0.11                                         | 0.14    |
| $\Sigma 3(1\bar{1}\bar{2})$ GB |                                      |           |                                              |         |
| O                              | 2.01/1.97                            | 2.07/1.97 | 0.17                                         | 0.25    |
| S                              | 2.25/2.13                            | 2.27/2.13 | 0.10                                         | 0.14    |
| Cl                             | 2.19/2.14                            | 2.2/2.15  | 0.11                                         | 0.12    |
| $\Sigma 5(0\bar{2}1)$ GB       |                                      |           |                                              |         |
| O                              | 2.08/1.99                            | 2.13/1.74 | 0.12                                         | 0.26    |
| S                              | 2.14/2.06                            | 2.29/1.92 | 0.08                                         | 0.21    |
| Cl                             | 2.15/2.01                            | 2.25/1.96 | 0.08                                         | 0.18    |

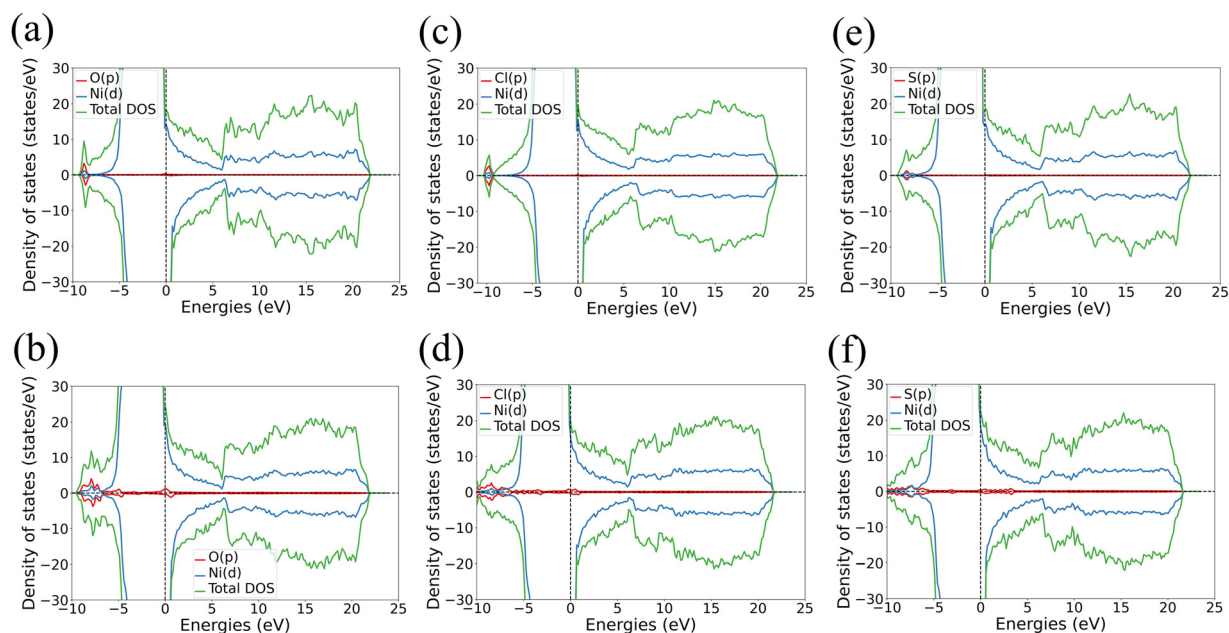
Cl atoms and the neighbouring Ni atoms at the interface of the  $\Sigma 3(111)$ ,  $\Sigma 3(1\bar{1}\bar{2})$ , and  $\Sigma 5(0\bar{2}1)$  GBs. It is evident that, although the number of electrons gained by impurity atoms slightly decreases with the increase in the impurity concentration, the average amount of the charge transferred from the Ni atoms to the impurity atoms in the GB increases. This confirms the charge redistribution between impurity atoms and the formation of new impurity–metal bonds. A similar tendency is found in the case of the  $\Sigma 3(1\bar{1}\bar{2})$  and  $\Sigma 5(0\bar{2}1)$  GBs, which is shown and discussed in the SI (Fig. S1 and S2).

Fig. 6 shows PDOS plots for the  $\Sigma 3(111)$  GB in Ni with impurity O, Cl, and S atoms. The presence of one impurity atom of O, Cl, and S in the  $\Sigma 3(111)$  GB results in strong bonding between Ni atoms of the GB and impurity atoms, which is indicated by a significant overlap between p orbitals of impurity elements and d orbitals of Ni atoms. The strongest overlap with

the spin-state splitting at around  $-10$  eV is found in the case of the O impurity (Fig. 6a). This is in line with the charge transfer analysis, showing the strongest interaction of the Ni atoms with the O impurity atom. The Cl-p peak overlaps with Ni d-orbitals at around  $-10$  eV (Fig. 6c), and the slightly broadened S-p peak overlaps with Ni d-orbitals at around  $-9$  eV (Fig. 6e). Fig. 6b, d and f, respectively, show that with an increase in the concentration of the O, Cl, and S impurities, a significant broadening and degeneracy of the impurity p states is observed. Therefore, the number of new impurity–metal bonds formed at the GB interface increases with the increase in the impurity concentration. A similar tendency is found in the case of the  $\Sigma 3(1\bar{1}\bar{2})$  and  $\Sigma 5(0\bar{2}1)$  GBs, which is shown and discussed in the SI (Fig. S3 and S4).

In the case of substitutional impurities, the strengthening/embrittlement energy of the GB consists of both mechanical and chemical contributions.<sup>16</sup> When an impurity atom segregates to a GB, it can displace the surrounding host atoms, creating mechanical changes, while the formation of new host–impurity bonds and also changes in the strength of nearby host–host bonds lead to a chemical change.

According to the charge transfer analysis above, O has high charge transfer to Ni atoms in the GB due to its high electronegativity, while the atomic radius of O is the smallest, thus mechanical deformation induced by O is low. Table 3 summarizes the data on the average/shortest Ni–O bond length and the average charge lost by the Ni atom in the GB. In the case of O impurities, the average and the shortest Ni–O bond lengths do not change remarkably with increasing impurity concentration, whereas the average charge transfer from Ni atoms to O atoms increases significantly. This is in line with our data in Table S1, which suggest comparably little change in the average



**Fig. 6** PDOS diagrams for the  $\Sigma 3(111)$  GB in Ni (grey spheres) with 1 (a) and 4 (b) impurity O (red spheres) atoms, with 1 (c) and 4 (d) impurity Cl (green spheres) atoms, and with 1 (e) and 4 (f) impurity S (yellow spheres) atoms.

distance between Ni atoms in the GB interface upon O intercalation. Fig. S5 and Table S2 present the mechanical contribution of impurities to the  $\Sigma 3(111)$  GB strength in Ni, which is defined as the difference between the energy of the pristine GB and the energy of the disordered GB after removal of impurity atoms. It is indeed confirmed that  $\gamma^{\text{dis}}$  in the case of O is higher than in the cases of Cl and S (Fig. S5a), however, with increasing O impurity concentration, the mechanical contribution increases compared to the chemical one from  $\sim 50\%$  to  $\sim 85\%$  (Fig. S5b). This is also consistent with literature, where the mechanical contribution for the interstitial O of about 50% of the total GB strength is reported.<sup>21</sup> Thus, in the case of O, the chemical and mechanical contributions are equally important.

Chlorine impurities introduce serious disordering in the GB, since the average distance between Ni atoms in the GB interface increases (Table S1). Table 3 shows that the average and the shortest Ni–Cl bond lengths as well as the average charge transfer from Ni atoms to Cl atoms do not change significantly. At the same time, Cl atoms gain low numbers of electrons from Ni atoms compared to O and S, as shown above. The calculated  $\gamma^{\text{dis}}$  in the case of Cl has the lowest values compared to the cases of O and S (Fig. S5a), decreasing to negative values with increasing impurity concentration, indicating the significance of the mechanical contribution. On the other hand, the mechanical contribution of Cl impurities varies significantly depending on the impurity concentration (Fig. S5b). This suggests that there is a competition between the mechanical and chemical contributions. Indeed, it has been experimentally observed that Cl ions accumulate at the Ni interface, forming stable compounds with metal ions and causing lattice expansion and structural inhomogeneity.<sup>9,54</sup>

In the case of sulfur,  $\text{NiS}_x$  or  $\text{Ni}_3\text{S}_2$  inclusions can be found in the GBs in Ni.<sup>55–57</sup> These inclusions can cause the mechanical stresses, hence spontaneous materials breakage. However, previous theoretical studies have shown that S impurities lead to the embrittlement effect mainly through the chemical contribution,<sup>16</sup> while the mechanical contribution is secondary. This is in line with our data in Table S1, which suggest comparably small changes in the average distance between Ni atoms in the GB interface upon S intercalation. Table 3 also shows that the average and the shortest Ni–S bond lengths do not change significantly, while a moderate increase in the average charge transfer from Ni atoms to S atoms is found when the impurity concentration increases. The calculated  $\gamma^{\text{dis}}$  values remain positive in the case of S (Fig. S5a). Although the mechanical contribution is valuable at low S impurity concentration, it decreases drastically with increasing impurity concentration (Fig. S5b). Therefore, the chemical effect, which includes formation of new Ni–S bonds and weakening the nearby Ni–Ni bonds, plays a more important role in GB strength degradation in that case.

## Conclusions

In conclusion, a detailed and systematic first-principles study is conducted to examine the influence of the O-, Cl-, and S-rich environment on the GB strength in the model Ni system. Various concentrations of O, Cl, and S impurities in the GB and

three representative Ni GBs, the  $\Sigma 3(111)$  coherent twin,  $\Sigma 3(112)$  incoherent twin, and high-angle  $\Sigma 5(0\bar{2}1)$  GBs, are considered. It is shown that the surface containing the  $\Sigma 3(1\bar{1}\bar{2})$  GB has the highest  $\gamma_{\text{FS}}$  value of  $2.65 \text{ J m}^{-2}$ , which makes it active towards corrosive environmental components. On the other hand, the  $\Sigma 3(111)$  GB has the lowest  $\gamma_{\text{GB}}$  value of  $\sim 0.10 \text{ J m}^{-2}$  and the highest  $W_{\text{sep}} = 3.67 \text{ J m}^{-2}$ , being the most favorable GB configuration in Ni. Furthermore, it is found that impurities, which segregate at the GBs, alter their composition and potentially decrease  $\gamma^{\text{imp}}$ . In most cases, the highest  $\gamma^{\text{imp}}$  is found when O segregates in the GBs, while the lowest  $\gamma^{\text{imp}}$  is associated with the Cl segregation. This suggests that even a low concentration of Cl atoms at the  $\Sigma 3(111)$  GB of Ni can significantly reduce its strength, leading to material degradation. Notably, both mechanical and chemical contributions are found to be remarkable in the case of O and Cl. The mechanical contribution increases compared to the chemical one from  $\sim 50\%$  to  $\sim 85\%$ , with increasing O impurity concentration. In the case of Cl impurities, a competition is found between the mechanical and chemical contributions depending on the impurity concentration.

The findings provided in this study enable atomistic-level understanding of corrosive impurities' impact on the GB strength in Ni and Ni-based alloys. This provides useful knowledge on the corrosion resistance of Ni and Ni-based alloys to corrosive impurities and supplies new methods of surface treatments or coatings of Ni and Ni-based alloys with some guidance on the development of more stable and effective barriers against corrosive agents, which is crucial for achieving the desired properties of these alloys during their production and for their technological applications.

Current results can be further expanded by studies on the co-segregation of impurities to GBs. Since GBs are usually exposed to an environment with different kinds of inherent impurities, these impurities often co-segregate in GBs, influencing their properties and dynamic behaviors.<sup>58</sup> Therefore, the coupling effects of different elements on the GB energy, strength, charge distribution and bond length may need to be considered in future studies.

## Conflicts of interest

There are no conflicts to declare.

## Data availability

The authors confirm that the data supporting the findings of this study are available within the article and/or its supplementary information (SI). Supplementary information is available. See DOI: <https://doi.org/10.1039/d5cp03364k>.

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